

THE AVERAGING METHOD IN OPTIMAL CONTROL PROBLEMS*

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(Received 28 March 1974)

THE PROBLEM of optimal control of an essentially non-linear system, containing a small parameter and reducible to a variable-phase system, is considered. It is assumed that the frequency of the system depends on slowly varying variables, and that the control functions appear in small perturbing terms, so that the system is weakly controlled. The averaging method is used to construct the solution of the optimal control problem for a long time interval. Examples are discussed.

1. Formulation of the problem and the maximum principle

Let the motion of the object be described by a system of differential equations with initial conditions:

$$\begin{aligned}\dot{x} &= \epsilon X(x, y, u, \epsilon), \quad x(t_0) = x^0, \\ \dot{y} &= \omega(x) + \epsilon Y(x, y, u, \epsilon), \quad y(t_0) = y^0.\end{aligned}\tag{1.1}$$

Here, x is an n -dimensional "slow" vector, y is the scalar "fast" phase, u is the m -dimensional control vector, ϵ is a small parameter, and t_0, x^0, y^0 are the initial data. The frequency $\omega(x) \geq \omega_0 > 0$ is assumed to be a fairly smooth function, essentially dependent on x , while X and Y are assumed to be periodic in y , with period 2π .

By analogy with the uncontrolled systems of a similar type, it seems natural to refer to Eqs. (1.1) as a variable-phase controlled system. A wide class of non-linear oscillatory and variable-phase systems containing a small parameter is known [1-3] to be reducible to Eqs. (1.1).

We pose the following optimal control problem for Eqs. (1.1): to find the piecewise continuous optimal control $u(t)$, which is unconstrained, and the corresponding optimal trajectory $x(t), y(t)$ in the interval $[t_0, T]$, for which the functional

$$J = g(x(T), \epsilon), \quad T - t_0 \sim 1/\epsilon.\tag{1.2}$$

is minimized. Here, T is the given instant of terminating the control process, and g is a fairly smooth function.

Notice that the functional

$$J = g_1(x(T), \epsilon) + \epsilon \int_{t_0}^T g_2(x, y, u, \epsilon) dt,\tag{1.3}$$

*Zh. vychisl. Mat. mat. Fiz., 15, 4, 869-882, 1975.

where g_2 is a periodic function in y , is reducible to the form (1.2).

We shall assume that the optimal control problem with the functional (1.2) has a unique solution in the absence of constraints on the control functions (for any sufficiently small $\epsilon \neq 0$). The optimal control and trajectory will then satisfy the maximum principle [4].

We denote by p the n -dimensional vector of variables containing the vector x , and by q , the scalar conjugate variable, corresponding to the y coordinate, We form the Hamiltonian function

$$H(x, y, p, q, u, \epsilon) = \epsilon(pX) + \epsilon qY + q\omega. \quad (1.4)$$

We write the conjugate system

$$\dot{p} = -\epsilon \frac{\partial(pX)}{\partial x} - \epsilon q \frac{\partial Y}{\partial x} - q \frac{\partial \omega}{\partial x}, \quad \dot{q} = -\epsilon \frac{\partial(pX)}{\partial y} - \epsilon q \frac{\partial Y}{\partial y} \quad (1.5)$$

and the transversality conditions

$$p(T) = -\frac{\partial g(x(T), \epsilon)}{\partial x}, \quad q(T) = 0. \quad (1.6)$$

Under these assumptions, the required optimal control maximizes the function (1.4) with respect to u for all t of $[t_0, T]$. Since there are no constraints on u , the condition for maximization of H with respect to u gives

$$\frac{\partial H}{\partial u_i} = \epsilon \left(p \frac{\partial X}{\partial u_i} \right) + \epsilon q \frac{\partial Y}{\partial u_i} = 0, \quad i = 1, 2, \dots, m, \quad (1.7)$$

where $u_1, \dots, u_m$ are the components of the vector u . Let Eqs. (1.7) be uniquely solvable with respect to the components of u . We can then obtain from (1.7) the relation

$$u = V(x, y, p, q, \epsilon), \quad (1.8)$$

where V is periodic in y .

Below we use the averaging method [1–3] to examine the asymptotic solution of the resulting boundary value problem (1.1), (1.5), (1.6), (1.8) in the first approximation with respect to the small parameter ϵ . Since the existence of a solution of the problem (1.1), (1.2) is assumed, the solution of the boundary value problem exists. Notice that perturbation theory methods were developed in [5, 6] for solving approximately optimal control problems. Weakly controlled optimal systems were investigated in [7] in a limited interval of time as $\epsilon \rightarrow 0$; controlled systems in the standard form [1, 3] were investigated by the averaging method in [2]; and systems of the type (1.1) with $\omega = \text{const}$ (ω is a vector) were considered in [8].

2. Construction of the asymptotic solution

Consider the system of $2n + 2$ differentiable equations (1.1), (1.5), in which u is replaced in accordance with (1.8). Formally, this system is not one with a single variable phase [1–3], since the variable p is not slow: its rate of change does not vanish for $\epsilon = 0$ and is equal to $(-q\partial\omega/\partial x)$. However, by using the boundary condition $q(T) = 0$ of (1.6), our system can be reduced to the standard form of a system with a single variable phase y .

For this, notice that the initial controlled system (1.1) is autonomous, and hence the system (1.1), (1.5), (1.8) has the first integral

$$H^*(x, y, p, q, \varepsilon) \equiv H|_{u=v} = \varepsilon(pX^*) + \varepsilon qY^* + q\omega = C = \text{const.} \quad (2.1)$$

Here, the asterisk means that the argument u of the relevant functions is replaced by expression (1.8).

Along the trajectories of the system (1.1), (1.5), (1.8), the quantity H^* is equal to a constant C . Since the vector $x(T)$ is assumed to be bounded, the vector $p(T)$, equal to $-g_x'(x(T), \varepsilon)$ (see (1.6)), will also be bounded. It follows from the boundary condition (1.6) that $C = \varepsilon(pX^*)_T$, i.e., $C \equiv \varepsilon h$, where $h \sim 1$ and

$$h = (pX^*)_T, \quad (2.2)$$

and h is bounded as $\varepsilon \rightarrow 0$. From (2.1),

$$q = \frac{\varepsilon}{\omega} [h - (pX^*) - qY^*], \quad \omega > 0, \quad (2.3)$$

i.e., formally, $q \sim \varepsilon$. Provided that the partial derivative of the right-hand side of (2.3) is bounded with respect to q for $q = \varepsilon = 0$, the equation is uniquely solvable for q :

$$q = \varepsilon \frac{h - (pX^*)}{\omega} + \varepsilon^2 q_2(h, x, y, p, \varepsilon), \quad X_0^* \equiv X^*(x, y, p, 0, 0). \quad (2.4)$$

Let us reduce the order of the system (1.1), (1.5), by substituting the expression (2.4) for q in it, where h is a parameter; we get

$$\begin{aligned} \dot{x} &= \varepsilon X_0^*(x, y, p) + \varepsilon^2 X_2(h, x, y, p, \varepsilon), & x(t_0) &= x^0, \\ \dot{y} &= \omega(x) + \varepsilon Y_0^*(x, y, p) + \varepsilon^2 Y_2(h, x, y, p, \varepsilon), & y(t_0) &= y^0, \\ \dot{p} &= -\varepsilon \frac{\omega'(x)}{\omega(x)} [h - (pX_0^*)] - \varepsilon \frac{\partial}{\partial x} (X_0^* p) + \varepsilon^2 P_2(h, x, y, p, \varepsilon), \\ p(T) &= -g_x'(x(T), \varepsilon). \end{aligned} \quad (2.5)$$

Here, the function Y_0^* is defined in a similar way to X_0^* , while X_2 , Y_2 , and P_2 , like q_2 , are bounded functions of their arguments which are 2π -periodic in q .

The solution of the boundary value problem (2.5) will be a function of t , of the parameters h and ε , of the initial values x^0, y^0 , and t_0 , and of T . We assume that, given any fixed $h \sim 1$, the solution of problem (2.5) exists and is unique:

$$x = x(t, h, \varepsilon), \quad y = y(t, h, \varepsilon), \quad p = p(t, h, \varepsilon). \quad (2.6)$$

To find the solution of the initial boundary value problem (1.1), (1.5), (1.6), (1.8), we have to satisfy the boundary condition for q ; the latter is the defining equation for the parameter h . The equation is

$$h = (p(T, h, \varepsilon)X^*(x(T, h, \varepsilon), y(T, h, \varepsilon), p(T, h, \varepsilon), 0, \varepsilon)), \quad (2.7)$$

and, in view of our assumption that the solution of the optimal problem (1.1), (1.2) exists, it has a solution in the real domain. If the root $h = h^*$ of the transcendental equation (2.7), which is bounded as $\varepsilon \rightarrow 0$, can be found, and we substitute it in (2.6), (2.4), in which, in turn, we substitute

the values obtained for the functions (2.6), we finally obtain the required solution of the problem (1.1), (1.5), (1.6), (1.8).

In view of the form of the right-hand side of (2.7), the parameter h will not in general be uniquely defined, since we have on the right the oscillating function h with frequency of the order of $1/\epsilon$, and amplitude of oscillation of the order of unity. In fact, by differentiating with respect to h , we can estimate the derivative $\partial y/\partial h$ at the instant T . Using (2.5) and the non-linearity of the system ($\omega'(x) \sim 1$), we obtain the following estimate in the principal term:

$$\frac{\partial y(T, h, \epsilon)}{\partial h} = \frac{\partial}{\partial h} \int_{t_0}^T \dot{y} dt \sim \frac{\partial}{\partial h} \int_{t_0}^T \omega(x) dt = \int_{t_0}^T \omega'(x) \frac{\partial x}{\partial h} dt \sim T \sim \frac{1}{\epsilon}.$$

Here we have used the equation $\partial y(t^0, h, \epsilon)/\partial h = \partial y^0/\partial h = 0$. On the other hand, the right-hand side of Eq. (2.7) is 2π -periodic in $y(T, h, \epsilon)$. This equation therefore has, in general, several roots, of the order $[1/\epsilon]$ for sufficiently small $\epsilon > 0$. From the admissible roots $\{h^*\}$ of Eq. (2.7) we obviously need to choose the one which minimizes the functional (1.2):

$$J(h) = g(x(T, h, \epsilon), \epsilon), \quad J^* = \min g(x(T, h, \epsilon), \epsilon) \quad (2.8)$$

with respect to $h \in \{h^*\}$, where $\{h^*\}$ is the set of roots of Eq. (2.7), the distance between adjacent roots being of order ϵ .

The solution of the optimal control problem (1.1), (1.2) is then defined by the functions (2.6), (2.4), (2.8), (1.8), taken at the optimal value of the parameter h^* .

Let us turn to obtaining the approximate solution of the system (2.5), which, given h , is the standard system with a single variable phase y . The boundary value problem of the first approximation is

$$\begin{aligned} \dot{\xi} &= \epsilon f(\xi, \psi), & \xi(t_0) &= x^0, & \dot{\psi} &= -\epsilon \frac{\omega'(\xi)}{\omega(\xi)} [h - (f\psi)] - \epsilon \frac{\partial}{\partial \xi}(f\psi), \\ \psi(T) &= -g_0'(\xi(T)), & \eta &= \omega(\xi), & \eta(t_0) &= y^0. \end{aligned} \quad (2.9)$$

Here,

$$f(\xi, \psi) = \frac{1}{2\pi} \int_0^{2\pi} X_0^*(\xi, y, \psi) dy, \quad g_0(\xi) = g(\xi, 0),$$

and the variables x, p, y are replaced by the corresponding averaged values ξ, ψ, η .

We know [1, 3] that the solution of the averaged system (2.9) is close to the solution of the initial system (2.5) in the following sense. Suppose we are given the system (2.5) with the parameter h and the initial conditions $x(t_0) = x^0$, $y(t_0) = y^0$, $p(t_0) = p^0$. Then, under extremely general assumptions, we have

$$|x - \xi| \leq \alpha \epsilon, \quad |p - \psi| \leq \alpha \epsilon, \quad |y - \eta| \leq \alpha, \quad \alpha = \text{const} < \infty, \quad (2.10)$$

for all $t_0 \leq t \leq T$, $T \sim 1/\epsilon$ for sufficiently small $\epsilon > 0$, provided that

$$|p^0 - \psi^0| \leq \beta \epsilon, \quad |h - k| \leq \beta \epsilon, \quad \beta = \text{const} < \infty,$$

where ξ, ψ, η is the solution of the averaged system (2.9) with initial conditions $\xi(t_0) = x^0, \psi(t_0) = \psi^0, \eta(t_0) = y^0$, and with $h = k$.

In view of our assumptions, the initial condition p^0 for the variable p in the initial boundary value problem (2.5) is uniquely determined by the quantities x^0, y^0, h , and ϵ , while the initial condition ψ^0 of the averaged system (2.9) is determined by x^0 and k . We shall now assume that the averaged boundary value problem (2.9) has a unique solution for every fixed value of the parameter, equal to k ; this solution has the following stability property. Suppose that, when small terms of order ϵ are added to the right-hand sides of the boundary conditions of (2.9), the solution of the boundary value problem at $t = T$ for the slow variables is changed by small quantities of the order ϵ throughout the time interval $[t_0, T]$. In this case, we can regard the problem (2.9) as a first approximation problem, since the inequalities (2.10) also hold here. In fact, the functions $x(t)$ and $p(t)$ will differ by terms of the order ϵ from the functions $\xi_*(t)$ and $\psi_*(t)$, which satisfy the averaged system (2.9) and the same initial data at $t = t_0$ as the functions $x(t), p(t)$. At the instant T , therefore, $\xi_*(t), \psi_*(t)$ will satisfy the boundary conditions of (2.9) with an error of order ϵ , and in accordance with the assumed stability property, they differ from the solution of the boundary value problem (2.9) by terms of order ϵ . Hence the difference between the functions $x(t), p(t)$ and $\xi(t), \psi(t)$ will amount to a quantity of order ϵ , i.e., inequalities of the type (2.10) also hold for the solutions of problems (2.5) and (2.9). Consequently, once the solution of the averaged problem (2.9) has been constructed, we can assume that the approximate solution of the problem (1.1), (1.2) has been found with an error of order ϵ with respect to the slow variables and the functional, provided that the value of the parameter h^* , optimal in the sense of (2.8), is known with the same error.

Let us turn to finding the approximate optimal value of the parameter h . The system (2.9) has the integral

$$\frac{1}{\omega(\xi)} [h - (f(\xi, \psi)\psi)] = \Phi = \text{const}, \quad (2.11)$$

as may be verified by direct differentiation of (2.11) with respect to t , in the light of (2.9) and the equation $\partial(f\psi)/\partial\psi = f$, which follows from (1.7) and the notation for f .

We introduce into the system (2.9) the scalar parameter Φ in accordance with Eq. (2.11):

$$\frac{d\xi}{d\tau} = f(\xi, \psi), \quad \xi(\tau_0) = x^0, \quad \tau = \epsilon t, \quad \tau_0 = \epsilon t_0, \quad (2.12)$$

$$\begin{aligned} \frac{d\psi}{d\tau} &= -\omega'(\xi)\Phi - \frac{\partial}{\partial\xi}(f\psi), & \psi(\theta) &= -g_0'(\xi(\theta)), & \theta &= \epsilon T \sim 1, \\ d\eta/d\tau &= \omega(\xi)/\epsilon. \end{aligned}$$

Here, τ is slow time, and the prime denotes the derivative with respect to ξ . The first two equations of (2.12) form the Hamiltonian system with the Hamiltonian function

$$H^*(\xi, \psi) = (f(\xi, \psi)\psi) + \omega(\xi)\Phi = h = \text{const}, \quad (2.13)$$

which is the first integral of the system (here we have to bear in mind the identity $\partial(f\psi)/\partial\psi = f$).

The parameters h and Φ are connected by the relation (2.11), so that, instead of finding h , we can find Φ . To do this, we substitute the value (2.13) for h in the condition (2.7), while taking the functions ξ, ψ at the instant $\tau = \theta$. In the right-hand side of Eq. (2.7) we substitute

$$\begin{aligned} x(T, h, \epsilon) &= \xi(\theta, \Phi) + O(\epsilon), & p(T, h, \epsilon) &= \psi(\theta, \Phi) + O(\epsilon), \\ y(T, h, \epsilon) &= \frac{1}{\epsilon} \int_{\tau_0}^{\theta} \omega(\xi) d\tau + y^0 + O(1) = \frac{1}{\epsilon} \left[\int_{\tau_0}^{\theta} \omega(\xi) d\tau + O(\epsilon) \right]. \end{aligned} \quad (2.14)$$

Here, $\xi(\tau, \Phi)$, $\psi(\tau, \Phi)$ is the solution of the boundary value problem (2.12); it depends on τ and the parameter Φ . After the substitutions, and elementary transformation, condition (2.7) becomes

$$\Phi = F\left(\xi(\theta, \Phi), \psi(\theta, \Phi), \frac{1}{\epsilon} \left[\int_{\tau_0}^{\theta} \omega(\xi(\tau, \Phi)) d\tau + O(\epsilon) \right]\right) + O(\epsilon), \quad (2.15)$$

where we have introduced the notation

$$F(\xi, \psi, y) = \frac{1}{\omega(\xi)} \{ \psi[X_0^*(\xi, y, \psi)] - f(\xi, \psi) \}. \quad (2.16)$$

The function F defined by Eq. (2.16) is 2π -periodic with respect to its argument y , and has zero mean over a period in y . Let

$$\frac{d}{d\Phi} \left[\int_{\tau_0}^{\theta} \omega(\xi(\tau, \Phi)) d\tau \right] \Big|_{\Phi=0} = \gamma_0 \neq 0. \quad (2.17)$$

Then, for small ϵ , the right-hand side of (2.15) will be a rapidly oscillating function of Φ in a neighbourhood of $\Phi = 0$, with frequency of order γ_0/ϵ , amplitude of order unity, and zero mean. In an interval of Φ , including $\Phi = 0$, therefore, Eq. (2.15) will have many roots, differing by quantities of the order of ϵ . But, to solve our problem with an error of order ϵ , it is sufficient to find h , and hence also Φ , with an error of order ϵ . Hence any value of Φ from an interval including the point $\Phi = 0$ will be admissible from the point of view of satisfying the condition (2.15) with the requisite accuracy.

Hence, to find Φ , we turn to the condition for minimizing the functional J with respect to Φ (or h), but instead of the exact condition (2.8), we concern ourselves with minimization with respect to all Φ belonging to a certain interval. Obviously,

$$J(h) = J_0(\Phi) + O(\epsilon), \quad J_0(\Phi) = g_0(\xi(\theta, \Phi)). \quad (2.18)$$

Here, $J(h)$ is the exact value of the functional, corresponding to a certain h , and $J_0(\Phi)$ is the approximate value, for which the parameter Φ is connected with h by Eq. (2.11).

We shall show that, under certain conditions, the optimal value of Φ , minimizing $J_0(\Phi)$, is zero. For, the necessary condition for an extremum is

$$\frac{dJ_0(\Phi)}{d\Phi} = (g_0'(\xi(\theta, \Phi)) \xi_{\Phi}(\theta, \Phi)) = 0, \quad \xi_{\Phi} = \partial \xi / \partial \Phi, \quad g_0' = \partial g_0 / \partial \xi. \quad (2.19)$$

We will transform Eq. (2.19). Recalling the boundary condition (2.12) for ψ and the fact that $\xi_{\Phi}(\tau_0, \Phi) = dx^0/d\Phi = 0$ inasmuch as the solution depends continuously on the parameter Φ , we obtain

$$\frac{dJ_0}{d\Phi} = -(\psi(\theta, \Phi) \xi_{\Phi}(\theta, \Phi)) = - \int_{\tau_0}^{\theta} \frac{d}{d\tau} (\psi(\tau, \Phi) \xi_{\Phi}(\tau, \Phi)) d\tau.$$

Differentiating the integrand with respect to τ and using the second equation of (2.12) for ψ , we arrive at

$$J_0'(\Phi) = \int_{\tau_0}^0 \left\{ \left[\omega_{\xi}' \Phi + \frac{\partial}{\partial \xi} (f\psi) \right] \xi_{\Phi} - \left(\psi \frac{d\xi_{\Phi}}{d\tau} \right) \right\} d\tau.$$

Using the equation in variations ($\partial f/\partial \xi$, $\partial f/\partial \psi$ are the matrices of partial derivatives)

$$\frac{d\xi_{\Phi}}{d\tau} = \frac{\partial f}{\partial \xi} \xi_{\Phi} + \frac{\partial f}{\partial \psi} \psi_{\Phi}, \quad \psi_{\Phi} = \frac{\partial \psi(\tau, \Phi)}{\partial \Phi},$$

which follows from Eq. (2.12), and also using the above-mentioned equation $\partial(f\psi)/\partial \psi = f$, we can transform the required derivative to

$$J_0'(\Phi) = \Phi \int_{\tau_0}^0 \omega'(\xi(\tau, \Phi)) \xi_{\Phi}(\tau, \Phi) d\tau = \Phi \frac{d}{d\Phi} \int_{\tau_0}^0 \omega(\xi(\tau, \Phi)) d\tau. \quad (2.20)$$

Comparing Eqs. (2.19) and (2.20), we see that $J_0(\Phi)$ seems likely to have an extremum at $\Phi=0$ in an ϵ -neighbourhood of this point, Eq. (2.15) has a root, in view of Eq. (2.17). Obviously, $\Phi=0$ will certainly be a local minimum point if (see Eq. (2.17))

$$\left. \frac{d^2 J_0}{d\Phi^2} \right|_{\Phi=0} = J_0''(0) = \frac{d}{d\Phi} \int_{\tau_0}^0 \omega(\xi(\tau, \Phi)) d\tau \Big|_{\Phi=0} = \gamma_0 > 0. \quad (2.21)$$

We find the condition for a global minimum of $J_0(\Phi)$. Integrating Eq. (2.20) by parts, we get

$$J_0(\Phi) = J_0(0) + \Phi \int_{\tau_0}^0 \omega(\xi(\tau, \Phi)) d\tau - \int_0^{\Phi} d\Phi_1 \int_{\tau_0}^0 \omega(\xi(\tau, \Phi_1)) d\tau. \quad (2.22)$$

It follows from this that $J_0(0)$ is a global minimum of the functional $J_0(\Phi)$ provided that, throughout the neighbourhood of the point $\Phi=0$ we have

$$\Phi \int_{\tau_0}^0 \omega(\xi(\tau, \Phi)) d\tau \geq \int_0^{\Phi} \left[\int_{\tau_0}^0 \omega(\xi(\tau, \Phi_1)) d\tau \right] d\Phi_1. \quad (2.23)$$

We thus arrive at the following procedure for constructing the approximate solution of the optimal control problem (1.1), (1.2). We first construct the single-parameter family of solutions $\xi(\tau, \Phi)$, $\psi(\tau, \Phi)$ of the boundary value problem (2.12), in which Φ is the parameter of the family, belonging to the neighbourhood of $\Phi=0$, and the corresponding family of functionals (2.18). Next we find the point Φ^* at which the functional $J_0(\Phi)$ has a minimum with respect to Φ . If conditions (2.21) and (2.23) hold, the optimal point is $\Phi^*=0$. The initial variables x , y , p , q , and the functional, are then found from the relations

$$\begin{aligned} x(t) &= \xi(\tau, \Phi^*) + O(\epsilon), & p(t) &= \psi(\tau, \Phi^*) + O(\epsilon), & q(t) &= O(\epsilon), \\ y &= \frac{1}{\epsilon} \int_{\tau_0}^{\tau} \omega(\xi(\tau_1, \Phi^*)) d\tau_1 + y^0 + O(1), \\ \tau &= \epsilon t, & J^* &= g_0(\xi(\theta, \Phi^*)) + O(\epsilon), & \theta &= \epsilon T. \end{aligned} \quad (2.24)$$

The optimal control u^* will be obtained with an error of order ϵ if we substitute $p = \psi(\tau, \Phi^*)$, $q = 0$, $\epsilon = 0$, in Eq. (1.8). The accuracy will not be reduced if we also put $x = \xi(\tau, \Phi^*)$, in Eq. (1.8); but it is not possible to replace the fast variable y in accordance with Eqs. (2.24) without loss of accuracy.

Notice that the solution of the averaged problem (2.12) is much simpler than that of the initial problem (1.1), (1.5), (1.6), (1.8). First, the system (2.12) for ξ and ψ contains $2n$ equations instead of $2n + 2$, and second, it contains no fast variables, so that we can eliminate ϵ by the time replacement $\tau = \epsilon t$, and reduce it to the solution of a boundary value problem in a short interval of slow time. In addition, by using the first integral (2.13) for the averaged canonical system (2.12), we can reduce its order, and in the case $n = 1$, reduce its solution to a quadrature.

3. Example

Consider the motion of a system with one degree of freedom, described by the equation

$$\ddot{x} = \epsilon [u + F(x) - 2\lambda \dot{x}], \quad x(0) = x_0, \quad \dot{x}(0) = v_0. \quad (3.1)$$

Here, x is the generalized coordinate, $\dot{x}$ is the velocity, $\epsilon \geq 0$ is a small parameter, ϵF is a small potential force, 2π -periodic in x and c , and with mean ϵF_0 over a period, while $2\epsilon\lambda$ is the coefficient of fluid friction.

Equation (3.1) can be used to describe a number of model examples of controlled motion in weak periodic fields, etc.

Substituting $\dot{x} = v$, we reduce Eq. (3.1) to a system of the type (1.1):

$$\dot{v} = \epsilon [u + F(x) - 2\lambda v], \quad v(0) = v_0, \quad \dot{x} = v, \quad x(0) = x_0. \quad (3.2)$$

Here, x is the fast, and v the slow, variable.

We shall seek the scalar control $u(t)$, which is optimal in the sense of a criterion of the type (1.3):

$$J = kv^2(T) + \epsilon \int_0^T u^2 dt, \quad k > 0, \quad T \sim \frac{1}{\epsilon}. \quad (3.3)$$

The averaged two-point problem corresponding to (3.2), (3.3), is described by equations of the type (2.12):

$$\begin{aligned} \frac{d\xi}{d\tau} &= \frac{\psi}{2} + F_0 - 2\lambda\xi, & \xi(0) &= v_0, & \tau &= \epsilon t, \\ \frac{d\psi}{d\tau} &= 2\lambda\psi - \Phi, & \psi(\theta) &= -2k\xi(\theta), & \theta &= \epsilon T. \end{aligned} \quad (3.4)$$

Here, ψ is the variable conjugate to ξ , Φ is a parameter, and $\omega(\xi) = \xi$ in the notation of Sections 1 and 2. Solving the linear boundary value problem (3.4), we obtain the following for $J_0''(\Phi)$ (see Eq. (2.21)):

$$\frac{1}{2} \frac{d^2 J_0(\Phi)}{d\Phi^2} = \frac{1}{2} \frac{d^2 J_0(0)}{d\Phi^2} = k\xi_1^2(\theta) + \frac{1}{4} \int_0^\theta \psi_1^2(\tau) d\tau, \quad (3.5)$$

where ξ , ψ_1 is the solution of the problem

$$\frac{d\xi_1}{d\tau} = \frac{\psi_1}{2} - 2\lambda\xi_1, \quad \xi_1(0) = 0, \quad \frac{d\psi_1}{d\tau} = 2\lambda\psi_1 - 1, \quad \psi_1(\theta) = -2k\xi_1(\theta).$$

Here we have used the facts that the solution of the problem (3.4) is linear in Φ and that the functional (3.3) is quadratic.

For $k > 0$, expression (3.5) is clearly strictly greater than zero, so that, in accordance with Section 2, we want to take $\Phi = 0$. Putting $\Phi = 0$, we obtain the optimal solution of the system (3.4):

$$\begin{aligned}\xi(\tau) &= \left[v_0 - \frac{F_0}{2\lambda} - \frac{\psi(\theta)}{8\lambda} e^{-2\lambda\theta} \right] e^{-2\lambda\tau} + \frac{F_0}{2\lambda} + \frac{\psi(\theta)}{8\lambda} e^{-2\lambda(\theta-\tau)}, \\ \psi(\tau) &= \psi(\theta) e^{-2\lambda(\theta-\tau)}.\end{aligned}\quad (3.6)$$

Here,

$$\begin{aligned}\psi(\theta) &= -2k\xi(\theta), \\ \xi(\theta) &= \left[v_0 e^{-2\lambda\theta} + \frac{F_0}{2\lambda} (1 - e^{-2\lambda\theta}) \right] \left[1 + \frac{k}{4\lambda} (1 - e^{-4\lambda\theta}) \right]^{-1}.\end{aligned}\quad (3.7)$$

The approximate values of the optimal control and of the minimum of the functional (3.3) are equal to

$$V = \psi(\tau)/2, \quad J_0(0) = \xi^2(\theta) k \left[1 + \frac{k}{4\lambda} (1 - e^{-4\lambda\theta}) \right]. \quad (3.8)$$

As a result, Eqs. (3.6)–(3.8) give the approximate solution of the optimal control problem (3.2), (3.3).

For clarity, suppose that $v_0 > 0$; it then follows from Eqs. (3.7) that, for $F_0 < 0$, a quantity θ^* exists, independent of k , such that $\xi(\theta^*) = 0$. This value of θ^* is

$$\theta^* = \frac{1}{2\lambda} \ln \left(1 - \frac{2\lambda v_0}{F} \right).$$

Since the averaging method is applicable under the assumption that $\omega(\xi) = \xi > 0$, we have to take $\theta < \theta^*$ (for the case $v_0 F_0 < 0$).

Further, from Eqs. (3.7) and (3.8), we can arrange by increasing $k > 0$, while keeping the other parameters fixed, that the value of $\xi^2(\theta)$ becomes arbitrarily small, though the value of the functional J_0 still remains bounded.

The problem with the functional (3.3) and $k > 0$ is an optimal braking problem. Notice that the optimal acceleration problem in the sense of the criterion (3.3) can also be solved with the aid of Eqs. (3.6)–(3.8). On varying k from zero to

$$k^* = -4\lambda(1 - e^{-4\lambda\theta})^{-1} < 0,$$

we can arrange for $\xi^2(\theta)$ to be as large as desired. Using Eq. (3.5), we can show that, with $k > k^*$, we have $J_0''(\Phi) = J_0''(0) > 0$, i.e., $\Phi^* = 0$ is the minimum point. Further reduction in k leads to the reverse inequality $J_0'' < 0$. Hence, by varying k , we can arrange both for "total" stopping $\xi(\theta) = 0$, $k = +\infty$, and for "total" acceleration $\xi^2(\theta) = \infty$, $k = k^*$.

4. Optimal control of the motion of a pendulum

As an application of the averaging method to a problem with constraints on the control, we take the problem of the optimal control of the motion of a pendulum. Under the action of a controlling disturbance $\epsilon u(t)$ of bounded modulus, the equation of motion of the pendulum is

$$\ddot{\varphi} + \sin \varphi = \epsilon u, \quad |u(t)| \leq 1, \quad 0 \leq t \leq T \sim 1/\epsilon. \quad (4.1)$$

Here, φ is the angular deviation of the pendulum from the bottom stable position of equilibrium, and $\epsilon \geq 0$ is a small parameter. It is required to find the control $u(t)$, minimizing the total energy E_* of the pendulum at a given finite instant T . Notice that the problem of maximizing E_* can also be solved by means of the relations obtained below. The solution procedure will differ somewhat from the general scheme of Sections 1 and 2, since here there are constraints on the control, and some specific features of the problem can be utilized.

The energy of the pendulum is

$$E_* = \dot{\varphi}^2/2 - (1 + \cos \varphi) \quad (4.2)$$

and $E_* = \text{const}$ for $\epsilon = 0$. As the origin for measuring the energy we take the upper equilibrium position of the pendulum $\varphi = \pi$, so that, if $E_* > 0$ the pendulum performs rotations, while it performs oscillations if $E_* < 0$. Differentiating Eq. (4.2) and using (4.1), we get

$$dE_*/dt = \epsilon \dot{\varphi} u, \quad (4.3)$$

i.e., E_* is a slow variable. Introducing a fast variable (phase) y by the usual method (see [1–3]), we can obtain a system of equations of the form (1.1). We shall only require Eq. (4.3).

To a first approximation, we find, using the constraint (4.1), that $\dot{u} = \text{sign} [\dot{\varphi}(E_*, y) \psi(t)]$, where $\psi(t)$ is the averaged conjugate variable corresponding to E_* , in accordance with the notation of Section 2. We substitute the control obtained in (4.3) and perform the averaging in accordance with (2.9). We denote the first approximation for the energy E_* by the same letter E_* . The averaged control for E_* takes the form

$$\frac{dE_*}{dt} = \epsilon \frac{\text{sign } \psi(t)}{2\pi} \int_0^{2\pi} |\dot{\varphi}(E_*, y)| dy.$$

Since our problem is to minimize E_* at the instant T , it is clear from this equation that the best result can be achieved if $\psi(t) < 0$ throughout the interval $[0, T]$. It remains to see if the conjugate equation has a solution with this property. The corresponding conjugate equation (see Eq. (2.12)), along with the boundary condition, has the form

$$\frac{d\psi}{dt} = -\epsilon \frac{|\psi(t)|}{2\pi} \frac{\partial}{\partial E_*} \int_0^{2\pi} |\dot{\varphi}(E_*, y)| dy - \omega'(E_*) \Phi, \quad \psi(T) = -1, \quad (4.4)$$

where $\omega(E_*)$ is the frequency of oscillation or revolution, and Φ is an unknown parameter (see Section 2). If we take e.g., $\Phi = 0$, the function $\psi(t)$ will have the property $\psi(t) < 0$ for $0 \leq t \leq T$. For, $\psi(T) < 0$, and with $\Phi = 0$ there is no value of t in the interval for which the solution of Eq. (4.4) can vanish. Hence $\psi(t) < 0$, and the optimal control is, to a first approximation,

$$u = -\text{sign } \dot{\varphi}. \quad (4.5)$$

This gives the approximate synthesis of the optimal control of the system (4.1). We shall now find the asymptotic solution of the equation obtained by substituting Eq. (4.5) into Eq. (4.1):

$$\ddot{\varphi} + \sin \varphi = -\varepsilon \text{sign } \dot{\varphi}. \quad (4.6)$$

Notice that this equation describes the motion of a pendulum with dry friction. To construct the solution, we follow [9], where a similar equation was solved.

For $\varepsilon = 0$, Eq. (4.6) describes the free oscillations or revolutions. The solution in the oscillatory mode is given by the relations [10]

$$\begin{aligned} \varphi &= 2 \arcsin \{k_1 \operatorname{sn}[k_1(t+t_0), k_1]\}, & 0 \leq k_1 \leq 1, & E \leq 0, \\ 1 + E/2 &= k_1^2, & T_1 &= 4K(k_1), & \varphi_0 &= 2 \arcsin k_1. \end{aligned} \quad (4.7)$$

Here, t_0 is an arbitrary constant, k_1 is the modulus of the elliptic functions, connected with the total energy E_* of the pendulum, T_1 is the period of oscillation, φ_0 is the amplitude of oscillation, and $K(k)$ is the complete elliptic integral of the 1st kind. In the rotatory mode we have [10]

$$\begin{aligned} \varphi &= 2 \operatorname{am}[(t+t_0)/k_2, k_2], & 0 \leq k_2 \leq 1, & E > 0. \\ 1 + E/2 &= 1/k_2^2, & T_2 &= 2k_2 K(k_2), \end{aligned} \quad (4.8)$$

where T_2 is the period of rotation, and k_2 is the modulus of the elliptic functions. We use the subscript 1 to refer to oscillation, and 2 to rotation. Using Eqs. (4.3), (4.5), and (4.7), and averaging over a period, we obtain for the oscillatory mode

$$\begin{aligned} \frac{dE_*}{d\tau} &= -\frac{1}{T_1} \int_0^{T_1} |\dot{\varphi}| dt = -\frac{4}{T_1} \int_0^{T_1/4} \dot{\varphi}(t) dt = -\frac{4}{T_1} \varphi_0 = -\frac{2 \arcsin k_1}{K(k_1)} \\ \tau &= \varepsilon t. \end{aligned} \quad (4.9)$$

Here we have used the fact that $\varphi(t)$ of Eq. (4.7) is an even function. Transforming to the variable k_1 and taking $\tau = 0$ as the instant when rotation becomes oscillation, we obtain from Eq. (4.9)

$$\frac{dk_1^2}{d\tau} = -\frac{\arcsin k_1}{K(k_1)}, \quad k_1(0) = 1, \quad \tau \geq 0. \quad (4.10)$$

Similarly, using Eq. (4.8), we find for the rotatory mode that

$$\frac{dE_*}{d\tau} = -\frac{1}{T_2} \int_0^{T_2} |\dot{\varphi}| dt = -\frac{2\pi}{T_2} = -\frac{\pi}{k_2 K(k_2)}. \quad (4.11)$$

Transforming to the variable k_2 in accordance with Eq. (4.8), we get

$$\frac{dk_2^2}{d\tau} = \frac{\pi}{2} \frac{k_2^3}{K(k_2)}, \quad k_2(0) = 1, \quad \tau \leq 0. \quad (4.12)$$

Note that, when $\tau > 0$ oscillation occurs; rotation occurs if $\tau < 0$. We can solve Eq. (4.12) in terms of complete elliptic integrals. Putting $z = k_2^2$ and using the expression of [11], p. 174, we get

$$\tau = \frac{2}{\pi} \int_1^{k_1^2} \frac{K dz}{z^{3/2}} = - \frac{4}{\pi} \frac{E(k_2)}{z^{1/2}} \Big|_{z=1}^{z=k_2^2} = \frac{4}{\pi} \left[1 - \frac{E(k_2)}{k_2} \right], \quad \tau \leq 0. \quad (4.13)$$

Here, $E(k)$ is the complete elliptic integral of the 2nd kind, and $E(1) = 1$. Expression (4.13) gives the solution of Eq. (4.12).

Certain properties of the functions $k_1(\tau)$, $k_2(\tau)$, following from Eqs. (4.10), (4.12), and (4.13), may be noted. Both functions decrease monotonically as $|\tau|$ increases. For small τ , when k_1 is close to 1, we find the asymptotic form of $k_1(\tau)$:

$$\tau = - \frac{1}{\pi} (1 - k_1^2) \left(\ln \frac{16}{1 - k_1^2} + 1 \right). \quad (4.14)$$

Alternatively, transforming to the energy E_* in accordance with Eq. (4.7), we obtain

$$\tau = - \frac{E_*}{2\pi} \left(\ln \frac{32}{|E_*|} + 1 \right). \quad (4.15)$$

Equation (1.14) enables us to start from $\tau = 0$ when integrating (4.10) numerically. Notice that Eqs. (4.14) and (4.15) hold for the rotatory mode, on replacing k_1 by k_2 , τ by $-\tau$, and E_* by $-E_*$.

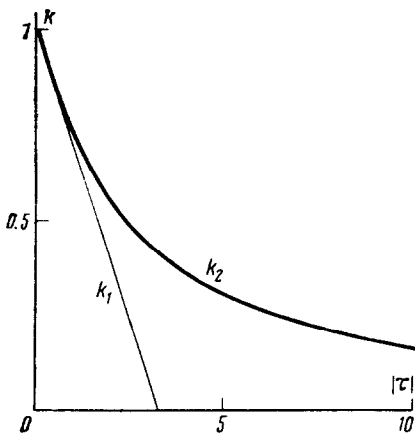


FIG. 1

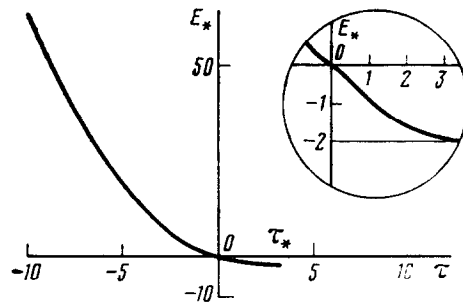


FIG. 2

Let us examine the asymptotic behaviour in another limiting case, when k_1 and k_2 are close to zero. Using the form of $K(k)$ for small k , we obtain from Eqs. (4.10) and (4.12)

$$k_1 = -(\tau - \tau_*)/\pi, \quad k_2 = -\frac{2}{\tau}, \quad |\tau| \rightarrow \infty. \quad (4.16)$$

It follows from the first of Eqs. (4.16) that k_1 vanishes for some finite value $\tau = \tau_*$, i.e., total stopping of the pendulum requires a finite time. The second of Eqs. (4.16) describes the decrease of k_2 as $\tau \rightarrow -\infty$.

The results of the numerical determination of $k_1(\tau)$ and $k_2(\tau)$ are given in Fig. 1. The curve $k_1(\tau)$ was found by integration of Eq. (4.10) in the light of the asymptotic form (4.14), and $k_2(\tau)$ by calculation based on Eq. (4.13). The value of the constant τ_* in Eq. (4.16) proved to be $\tau_* = 3.342$. This constant is equal to the dimensionless time needed to stop the pendulum in its lower equilibrium position, when it was initially at rest in the upper equilibrium position. In Fig. 2 we show in different scales the energy dependence $E_*(\tau)$ for the oscillatory mode ($\tau \geq 0$) and the rotatory mode ($\tau < 0$), obtained by calculating $k_1(\tau)$, $k_2(\tau)$ from Eqs. (4.7) and (4.8).

The dependence $E_*(\tau)$ is universal and enables various problems on minimization or maximization of the energy of a pendulum to be solved. Suppose that, at the initial instant, the pendulum energy is $E_* = E_0$, and that we want to construct an optimal control in an interval of length τ_0 , aimed at minimizing or maximizing the final value of the energy. For this, we find the point τ_1 on the graph of $E_*(\tau)$, corresponding to the value $E_*(\tau_1) = E_0$. Since $E_*(\tau)$ is monotonic, the point τ_1 is unique. Then, from the point τ_1 we mark off on the abscissa axis an interval of length τ_0 in the direction of increasing (decreasing) τ , for the minimization (maximization) problem. The values $E_*(\tau_1 \pm \tau_0)$ obtained give respectively the minimum and maximum energy at the end of the control interval.

If $E_*(\tau_1 + \tau_0) = -2$, τ_0 represents the optimal time needed to stop the pendulum with the initial energy $E_*(\tau_1) = E_0$. This follows from the fact that $E_* = -2$ corresponds to the stable position of equilibrium (see Eq. (4.2) with $\varphi = \dot{\varphi} = 0$). Hence the curve $E_*(\tau)$ also enables the time optimal problem to be solved approximately.

For comparison, we give below the optimal times corresponding to the exact and the approximate optimal control:

x	π	2.44	1.60	-0.15	-2.95	-4.76	-6.62
$\dot{x}$	0.00	1.66	3.52	5.00	3.20	4.95	6.00
T	8.06	8.13	8.33	8.64	9.27	9.67	9.96
T_0	7.47	7.58	8.02	8.36	9.04	9.47	9.72

The exact optimal time T was found by A. B. Doktorovich by constructing approximately the optimal trajectories for the system

$$\ddot{x} + \alpha \sin x = u, \quad |u| \leq 1, \quad x(T) = 2N\pi, \quad \dot{x}(T) = 0, \quad (4.17)$$

where α is the parameter and N is an arbitrary integer. The approximate value of T_0 was found for given x and $\dot{x}$ by calculations based on the curve $E_*(\tau)$. Comparing Eqs. (4.1) and (4.17), we have in the present case

$$E_* = \frac{1}{2\alpha} \dot{x}^2 - (1 + \cos x), \quad T_0 = \alpha^{1/2} \tau_0, \quad e = 1/\alpha,$$

where τ_0 is the dimensionless optimal time for the initial energy E_* . The results quoted refer to the case when $\alpha = 5$, i.e., $\epsilon = 0.2$.

The authors thank N. N. Bolotnik for assistance.

Translated by D. E. Brown

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THE INTERIOR POINT METHOD IN THE CONSTRAINED MULTIPLE MAXIMIN PROBLEM*

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(Received 13 July 1973; revised 1 October 1973)

THE APPLICATION of the method of penalty functions to maximin problems is discussed. Convergence theorems are proved for the interior point method whereby coupled constraints can be removed from the multiple maximin problem. In addition, a composite exterior-interior point method for use in the simple maximin problem is discussed.

Introduction

Convergence theorems for the exterior and interior point methods in problems of mathematical programming were proved in [1]. The application of methods of penalty functions to the minimax problem has so far been confined to the exterior point method (using the terminology of [1]). An attempt was made in [2] to extend the results of the exterior point method to minimax problems with coupled constraints.

In the present paper we prove theorems on the convergence of the interior point method (Section 1) for a quite general problem of finding a multiple maximin with coupled constraints:

*Zh. vychisl. Mat. mat. Fiz., 15, 4, 883–894, 1975.